Noâm Boussouf

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Academic Profile

Graduate researcher in Computer Science at Keio University (HVRL) and École Centrale de Nantes, with a strong theoretical foundation from PSL University (M2 IASD). First author of two papers in monocular 3D reconstruction and temporal video modeling (ACM MMSports '26, WACV '27 under review). Research aspirations center on developing Computer Vision methods for medical image analysis, with particular interests in unsupervised and representation learning for domain adaptation, anomaly detection, and segmentation, while enthusiastic about exploring the broader spectrum of Computer Vision challenges for medical applications.

Publications

Peer-Reviewed Conference Proceedings

- **Noâm Boussouf**, Adrien Maglo, Mohamed Chaouch, Paul Pham Dang, Hideo Saito.
  "*PoseConvSpot: Overcoming Domain Shifts in Precise Event Spotting for Sports Videos via Pose-Only Spatio-Temporal Regression*".
  In **Proceedings of the 9th ACM International Workshop on Multimedia Content Analysis in Sports (MMSports '26)**, in conjunction with ACM Multimedia, Rio de Janeiro, Brazil, 2026.
  Role: First Author. Project page: `noambsf.github.io/PoseConvSpot`

Manuscripts Under Review

- **Noâm Boussouf**, Adrien Maglo, Mohamed Chaouch, Hideo Saito.
  "*Volley3D: Unified 3D Full-Sequence Reconstruction from Monocular Broadcasts*".
  Under review at the **IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027)**, Main Conference (Applications Track), 2027.
  Role: First Author. Paper ID: #2401

Education

Keio University *Yokohama, Japan*
Master of Science in Computer Science (Research-Track) *Oct. 2025 – Present (Exp. Aug. 2027)*
Hyper Vision Research Laboratory (HVRL), advised by Prof. Hideo Saito.

Academic Focus: 100% research-intensive graduate curriculum focused on Computer Vision.
Cumulative GPA: [3.75 / 4.00] (1st year completed).
Graduate Coursework: Computer Vision, Advanced Signal Processing, Non-Linear Dynamics, Quantum Information Theory, Applied Statistical Analysis...
Admitted through a selective, competitive Double Degree program with École Centrale de Nantes.

PSL University (Dauphine, ENS Paris, Mines Paris) *Paris, France*
Master 2 IASD – M.Sc. in Artificial Intelligence, Systems, and Data *Sept. 2024 – Sept. 2025*
Ranked #1 in France, #24 Worldwide (QS World University Rankings), #33 THE World Rankings.

Highly selective, theoretical research master coordinated by Université Paris-Dauphine, École Normale Supérieure (ENS Paris) and Mines Paris.
Master Thesis: *"Day-to-Night Image Translation for Nighttime Detection and Image Restoration".*
Rank: 6th / 53 in a cohort composed of top French Grande École graduates (Polytechnique, ENS, Centrale Paris, Mines Paris...). Cumulative Grade: 15.1/20.
Core Curriculum: Learning Theory, Optimization, Non-Convex Inverse Problems, 3D Point Clouds, Generative AI, Optimal Transport, Reinforcement Learning...

École Centrale de Nantes *Nantes, France*
Diplôme d'Ingénieur (Equivalent to Master of Engineering / M.Eng.) *Sept. 2022 – Present (Dual Degree)*

Top-tier French Grande École – Specialization in Computer Science for Artificial Intelligence.

Year 2 (M1 equivalent): Foundations of Deep Learning, Reinforcement Learning, Computer Vision, Graph Theory – GPA: **4.00 / 4.00**.
Year 1 (B.S. equivalent): Advanced Applied Mathematics, Numerical Optimization, Probability & Statistics – GPA: **3.78 / 4.00**.
Academic Award: Recipient of the **Centrale Nantes Foundation Scholarship** (merit-based award for International studies).

Lycée Pierre de Fermat *Toulouse, France*
Classes Préparatoires aux Grandes Écoles (PCSI / PC)* *Sept. 2020 – July 2022*
Intensive 2-year undergraduate preparation in advanced Mathematics and Theoretical Physics for competitive national exams.
Cumulative GPA: **3.88 / 4.00**.

Research Experience

Hyper Vision Research Laboratory (HVRL), Keio University *Oct. 2025 – Present*
Graduate Research Assistant (Advised by Prof. Hideo Saito) *Yokohama, Japan*

- **Volley3D (WACV 2027 under review, First Author):** Developed a monocular 3D sports reconstruction framework from broadcast footage scenes, unifying physics-corrected player jump positioning with a context-aware 2D-to-3D ball lifting pipeline introducing a TCN-based trajectory segmenter, heuristic collision matching, and an aerodynamic solver anchored to reconstructed 3D hands.
- **PoseConvSpot (ACM MMSports '26):** Designed a pose-only spatio-temporal regression architecture (Strip Pooling, GRU, Soft Gamma Loss) for Precise Event Spotting (PES). Achieved a **+14.1% mAP gain** under extreme cross-arena domain shifts over state-of-the-art RGB models.

GoPro, PSL University *April 2025 – September 2025*
Computer Vision Research Intern – Master Thesis *Paris, France*

- Investigated **unsupervised Image-to-Image (I2I) translation** to synthesize realistic nighttime training data from daytime imagery, addressing extreme low-light data scarcity. Developed an enhanced CycleGAN incorporating structural priors (leveraging DINO) alongside an RGB-to-RAW unprocessing pipeline, allowing seamless adaptation of RAW-to-RAW and RGB-to-RGB **domain transfer** models.
- Formulated a dual-task evaluation protocol assessing both low-level restoration (denoising metrics) and high-level perception (object detection mAP) to assess synthetic domain transfer performances.

CEA LIST, Université Paris-Saclay *Apr. 2024 – Aug. 2024*
Computer Vision Research Intern (LVA Lab) *Palaiseau, France*

- Conducted a research project on single-view 3D reconstruction of collective sports scenes using camera calibration, multi-player tracking, and 3D human pose estimation.
- Formulated initial kinematic modeling combining action recognition and physics-based equations to resolve jump ambiguities in volleyball, establishing foundations for **MMSports '26** publication.

Nantes University Hospital (CHU de Nantes) *Oct. 2023 – Mar. 2024*
Machine Learning Research Project *Nantes, France*

- Project with the Nantes University Hospital aimed at predicting lung transplant rejection in patients using Machine Learning on clinical data, assisting a doctoral student in his thesis.

Skills & Languages

Deep Learning & Vision	PyTorch, CUDA C, OpenCV, TorchVision, NumPy, SciPy (Optimization)
Software & 3D Tools	Python, C/C++, Blender, Linux, Bash, Git, Slurm (HPC Clusters), LaTeX
Languages & Tests	**French** (Native), **English** (**IELTS: 7.5/9** – TOEIC: 925/990), **Japanese** (Conversational)